

Om Shah

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EDUCATION

MPSTME, NMIMS University

Bachelor of Technology in Computer Engineering; CGPA: 3.96/4.0

Mumbai, India

2022 – 2026

R.N. Podar School

12th CBSE: 97%

Mumbai, India

2022

PROFILE

Computer Engineering student with strong interest in machine learning and AI systems. Experience working with deep learning frameworks, building RAG pipelines, and implementing reinforcement learning algorithms for practical applications.

PROJECTS

Agflow: Visual AI Agent Orchestration Platform | *Python, React Flow, LangChain*

- Developed an open-source platform for constructing agentic artificial intelligence workflows through a node-based visual interface powered by React Flow. Integrated retrieval-augmented generation pipeline utilizing Supabase pgvector for document processing with automated chunking and embedding capabilities. System features include multi-model support for Groq and OpenAI, pre-configured tool nodes for web search and application programming interfaces, custom Python code execution via Monaco Editor, and cloud-based flow management with automatic saving functionality.

StudHelp: Multi-Document RAG Study Assistant | *Next.js, LangChain, Supabase*

- Engineered a full-stack retrieval-augmented generation application featuring Supabase authentication for secure user management and document privacy protection. Implemented multi-document conversational interface with context-aware responses and citation mechanisms, automated document summarization with key point extraction, interactive mind map generation capabilities, and quiz generation features. Constructed portable document format viewer with zoom controls, page navigation functionality, and annotation system enabling users to save notes directly on documents.

Knowledge Distillation in LLMs using Jensen-Shannon Divergence | *PyTorch*

- Implemented a novel Jensen-Shannon Divergence-based knowledge distillation framework for compressing language models, addressing the inherent limitations of Kullback-Leibler Divergence. Successfully distilled SmolLM2-135M to 90 million parameters, achieving a mean F1 score of 0.9125 compared to Kullback-Leibler Divergence's 0.9028 with 12.50% reduction in variance. Demonstrated 3.45% improvement in ROUGE scores and 20% acceleration in convergence with 23.3% reduction in final loss, empirically validating Jensen-Shannon Divergence's theoretical advantages in optimization stability and dark knowledge preservation.

Agentic RAG System for Research Paper Comprehension | *FAISS, LangChain, PyTorch*

- Constructed a locally-hosted retrieval-augmented generation system for Computer Vision and Pattern Recognition research papers with automated arXiv scraping pipeline for metadata and full-text extraction. Implemented FAISS vector database utilizing sentence-transformer embeddings for efficient retrieval and quantized low-rank adaptation-based fine-tuning of Qwen models for domain-specific adaptation. Developed LangChain agent with tool-calling capabilities for reasoning over academic literature, incorporating comprehensive evaluation framework using Hit Rate, Mean Reciprocal Rank, and ROUGE metrics.

Reinforcement Learning for Portfolio Management | *PyTorch, TensorFlow*

- Implemented and evaluated Deep Deterministic Policy Gradient, Proximal Policy Optimization, and Dynamic Embedding Reinforcement Learning algorithms for dynamic asset allocation on Indian equities utilizing daily market data spanning 2012 to 2023. Trained agents on multi-year National Stock Exchange data incorporating proportional transaction costs and realistic market friction modeling. Deep Deterministic Policy Gradient agent achieved superior performance with 723.5% total return and 1.78 Sharpe ratio, while Dynamic Embedding Reinforcement Learning with Wasserstein autoencoder state representation achieved 357.8% return, both substantially outperforming the passive benchmark's 185.4% return.

Neural Machine Translation with Transformer Architecture | *PyTorch, spaCy*

- Constructed a Spanish-to-English neural machine translation system utilizing Transformer architecture with multi-head self-attention mechanisms and sinusoidal positional encodings. Implemented advanced preprocessing pipeline combining spaCy lemmatization for linguistic normalization with SentencePiece Byte Pair Encoding tokenization (16,000 vocabulary size) to address out-of-vocabulary challenges. Trained on 500,000 sentence pairs from the Europarl v7 corpus, achieving validation perplexity of 4.613 and test bilingual evaluation understudy score of 12.41 with beam search decoding strategy.

Football Video Analysis System | *YOLOv8, OpenCV, Django*

- Developed a real-time tracking system for analyzing player movements, velocity metrics, and ball possession statistics utilizing computer vision techniques integrated with Django web interface.

Face Detection using Deep Learning | *TensorFlow, Keras*

- Constructed a convolutional neural network-based face detection system incorporating custom data generators and training procedures utilizing the WIDER FACE dataset.

Dynamic Chat Room Application | *MERN Stack, JWT*

- Engineered a full-stack chat application featuring JSON Web Token authentication, MongoDB for persistent data storage, and Express/Node.js backend infrastructure.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML, CSS, SQL
AI/ML: PyTorch, TensorFlow, LangChain, Transformers, FAISS, ChromaDB, OpenAI API, RAG Pipelines, Neural Networks, Reinforcement Learning
Web Development: ReactJS, Next.js, Node.js, Express, Flask, Django, MongoDB, Supabase, PostgreSQL
Tools: Git, Docker, Vercel, Render, spaCy, OpenCV

CERTIFICATIONS

Quantum Machine Learning <i>EduXLabs, IIT-Madras</i>	2025
Deep Learning Specialization <i>DeepLearning.AI</i>	2024
AWS Academy Cloud Foundations <i>Amazon Web Services</i>	2024
Google Data Analytics Professional Certificate <i>Google</i>	2024
Honours Program <i>Coursera</i>	2022 – 2026

LEADERSHIP & VOLUNTEERING

Technical Executive <i>MPSTME ACM Student Chapter</i>	2023 – 2024
Class Representative <i>BTech Computer Engineering, Division B</i>	2022 – Present
Computer Tutor <i>Underprivileged Children Education</i>	2023 – Present